Sanny Hong

AI Engineer · Production LLM & Agentic Systems · Evaluation · Cost & Latency · Computer Vision

synhong43@gmail.com · linkedin.com/in/sanny-hong-020781187 · github.com/HSanny · Singapore

INTRODUCTION

AI engineer who builds and runs LLM systems in production. In the technical department of an industrial-IoT company, I shipped provider-agnostic agentic assistants over a 19M-row database, cut tokens per question by 84%, and check answers against ground-truth queries on live data. I also ship computer-vision and time-series models that run on CPU-only hardware.

CORE SKILLS

LLM & Agents: OpenAI and Anthropic APIs, tool/function calling, agentic loops, context engineering, RAG and grounding, structured outputs, provider-agnostic model layer

Evaluation & Reliability: ground-truth evaluation, deterministic tests, error analysis, entity resolution, per-request cost caps, rate-limit (TPM) handling, structured run/step tracing

Machine Learning: PyTorch, OpenCV, segmentation, OCR, detection and tracking, anomaly detection (Isolation Forest, LOF, LSTM, S-H-ESD), time-series forecasting, CPU inference

Data & Engineering: Python, SQL, TypeScript; FastAPI, Django, React; PostgreSQL, SQL Server; Docker; ETL; CI/CD

EXPERIENCE

Software Engineer — Flotech Controls, Singapore *Dec 2024 – Present*

Regional process-control & instrumentation provider. Technical department: AI for public-sector water infrastructure (SCADA / industrial IoT).

End-to-end ownership: data processing and labelling, model development and fine-tuning, deployment, production integration (PostgreSQL / SQL Server, scheduling, alerting) and monitoring, with documentation for each system.

- **Agentic LLM assistants:** Shipped production LLM assistants over a 447-station, 19M-row time-series database: an agentic tool-calling loop over parameterised, read-only SQL, so the model can query but never modify production data.
- **Provider-agnostic model layer:** OpenAI and Anthropic behind one interface, including a compatibility shim for GPT-5 API changes, so models can be swapped without changing agent code.
- **Token & cost control:** Cut tokens per question (input + output) from ~233k to ~36k (−84%) through context engineering, tracked as cost per response, which kept traffic under provider rate limits. Enforced per-request USD cost caps in production.
- **Ground-truth evaluation:** Scripts check answers against direct SQL on live data. They caught a silent mis-join that attributed work orders to 1 site instead of 46; fixed with a schema-relationship context file and entity resolution. Every agent step and run is logged as JSONL traces.
- **Visual detection pipeline:** Fine-tuned DeepLabV3-MobileNetV3 (IoU 0.841) for flood-level detection and fused six detectors with OCR auto-calibration, cutting reading error from ~137px to ~1px.
- **Model re-engineering:** Re-engineered a Florence-2 + SAM2 GPU pipeline into a CPU-only OpenCV stack (<1 s/image) with the same accuracy on the same test set, plus auto-generated training labels.
- **Time-series anomaly detection:** Production anomaly detection on high-frequency water-quality data (Isolation Forest, LOF, LSTM, S-H-ESD) with ETL, scheduling and alerting; redesigned flat tables into a star schema for fast SCADA analytics.

Independent Projects & Further Study — Singapore *Dec 2022 – Nov 2024*

- Built an Amazon-style e-commerce web app with a content-based recommender, later rebuilt as the Forecasting Engine; admitted to SMU's Master of IT in Business (AI) on scholarship.

Software Engineer (Frontend Lead) — GienTech, Shanghai *Feb 2022 – Dec 2022*

Seconded to OCBC. China-based expert in digital-transformation strategy and solutions.

- Led front-end development of OCBC Malaysia's SIBS Connect financial-record system (1,000+ users) across the full lifecycle; mentored junior engineers; SonarQube + Jenkins CI/CD with 90%+ deployment success.
- Delivered an HR management system (React / TypeScript / Django) for 300+ staff on bi-weekly releases. During a JavaScript → TypeScript migration, consolidated duplicated logic into shared React contexts, cutting the codebase by ~40%.

PERSONAL PROJECTS

Agentic AI — Forecasting Engine *2023 – Present*

- Natural-language agent over a dataset-agnostic recommender and time-series demand forecaster. Retrieval grounds every answer in real catalogue items, and the agent refuses when nothing matches. Each answer carries its intent, tool and evidence; adapters for H&M and Amazon data; 12 deterministic tests. github.com/HSanny/forecasting-engine

Computer Vision — Motion Analyser *2026 – Present*

- Turns one camera into an event stream: detection and tracking feed line-crossing, zone-dwell and speed analytics. Analytics core unit-tested; end-to-end run on real footage in progress. github.com/HSanny/motion-analyser

EDUCATION

Master of IT in Business (Artificial Intelligence) — Singapore Management University *2024 – Present*

Credit: SMU MITB Scholarship

Bachelor of Information Technology — The Australian National University *2017 – 2022*

Majors: Computer Science & Mathematics · Specialisation: Intelligent Systems · Minor: Japanese Language & Culture

LANGUAGE

Languages: English (full professional) · Mandarin, Cantonese (native) · Japanese (conversational)